

language, enabling the AI/ML to automatically set the parameters of the solvers and run simulations for all possible design choices to provide the best possible product choice.

Our long-term technology strategy shown in the figure on the previous page reveals how we are planning to solve this problem for our customers over the next 20 years.

By 2041, computers won't simply continue to march to the productivity increases inherent in Moore's Law. Simulation software will take advantage of the

software via increased automation and AR that will make software easier to use.

In addition to enabling the practical use of finer meshes for greater accuracy, new mesh innovations will flourish in such an environment. For example, isogeometric analysis meshing will natively bring FEA into conventional computer-aided design (CAD) tools to allow responsive simulation for complex models.

As software gets smarter and more responsive thanks to more

parts, MBSE will support large-scale digital mission engineering and end-to-end engineering lifecycle management that will have far-reaching effects on design for manufacturing, sustainable material choices and knowledge transfer.

In 2041 simulation will be faster, accurate and ubiquitous. It will extend beyond parts, assemblies and systems to encompass multiphysics across multiple domains. While the fulfillment of those predictions alone has world-altering repercussions, it seems a bit safe. Besides, what will all the simulation experts be doing if novices and AI are running simulations? They'll bypass what AI can reveal based on existing product data and focus on the atoms that make up the molecules that make up the products.

Think about it: Someone has characterized materials and found Young's modulus so that we can simulate the effects of stress and strain. But inside those materials are atoms or molecules. If we could go there, we could simulate without limitation. Truly optimized creations – based on the best materials at a molecular level – would spring forth regardless of our preconceived notions about what a product should look like. It would look like it should.

For example: Instead of designing a CAD model and performing structural analysis to simulate it to see if a soda can design works, we would specify – via natural language instead of a CAD user interface – that we need a container to hold 12 ounces of X and then the software would generatively design and simulate options at a multi-scale level, down to effervescence of the soda.

Now imagine if the soda can was a life-extending pharmaceutical and the requirements were based on an individual patient's biological vitals. The result would be the perfect pill, additively manufactured on-demand for each patient.

Just imagine the possibilities, then simulate them and then create them. Will we be there in 20 years? With simulation, all things are possible. 



Smart cities will rely on simulation for everything from setting land and air traffic patterns to predicting energy use.

speed increases that exascale and quantum computing bring to the entire hardware stack, from processors to memory to interconnects. Engineers won't need to choose between speed and accuracy. Many simulations will solve in near real-time, enabling on-the-fly exploration of design modifications. The increased compute power will also provide answers to complex problems that would be impractical to fully simulate today.

Computing innovations will spin off speed boosts at all levels: cloud computing, on premises HPC and the edge where connected devices will collect data that will be used to inform digital twins and help engineers optimize the next generation of products. All of that data and more will be needed to train AI algorithms that will fundamentally alter simulation

advanced HPC and AI, it will also visually guide experts and novices alike through simulation setup, optimization and solving. This will democratize simulation throughout product design and development and beyond, giving novices the confidence to use simulation like they use PowerPoint today. Interacting with AR-enabled digital twins will be the status quo for virtual testing.

Simulation: Grand to Granular Scale

The increased speed, accuracy and ease of use will enable model-based system engineering (MBSE) to expand its foothold in many verticals. MBSE supports system requirements from the earliest conceptual stage to connect people, hardware and software design processes and workflows at the system level, rather than treating them as individual